Ladder graph

In the mathematical field of graph theory, the **ladder graph** L_n is a planar, undirected graph with $2n$ vertices and $3n - 2$ edges.[1]

The ladder graph can be obtained as the Cartesian product of two path graphs, one of which has only one edge: $L_{n,1} = P_n \times P_2$.[2][3]

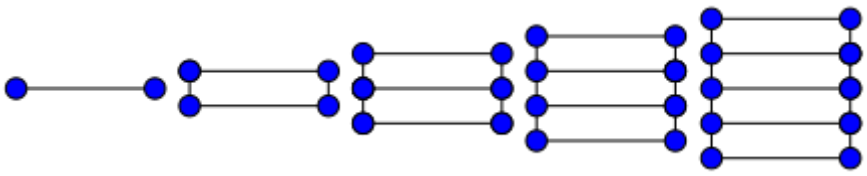
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Properties

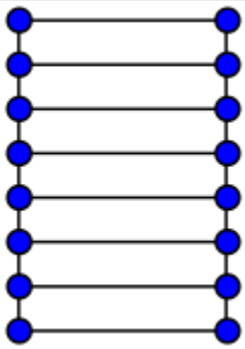
By construction, the ladder graph L_n is isomorphic to the grid graph $G_{2,n}$ and looks like a ladder with n rungs. It is Hamiltonian with girth 4 (if $n>1$) and chromatic index 3 (if $n>2$).

The chromatic number of the ladder graph is 2 and its chromatic polynomial is $(x-1)x(x^2-3x+3)^{(n-1)}$.



The ladder graphs L_1, L_2, L_3, L_4 and L_5.

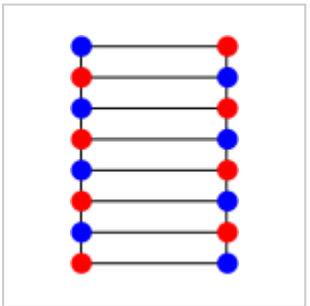
Ladder graph



The ladder graph L_8.

Vertices	$2n$
Edges	$3n - 2$
Chromatic number	2
Chromatic index	3 for $n > 2$ 2 for $n = 2$ 1 for $n = 1$
Properties	Unit distance Hamiltonian Planar Bipartite
Notation	L_n

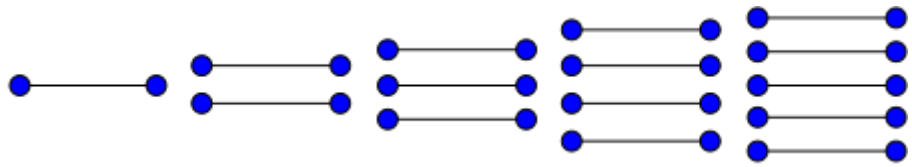
Table of graphs and parameters



The chromatic number of the ladder graph is 2.

Ladder rung graph

Sometimes the term "ladder graph" is used for the $n \times P_2$ **ladder rung graph**, which is the graph union of n copies of the path graph P_2.



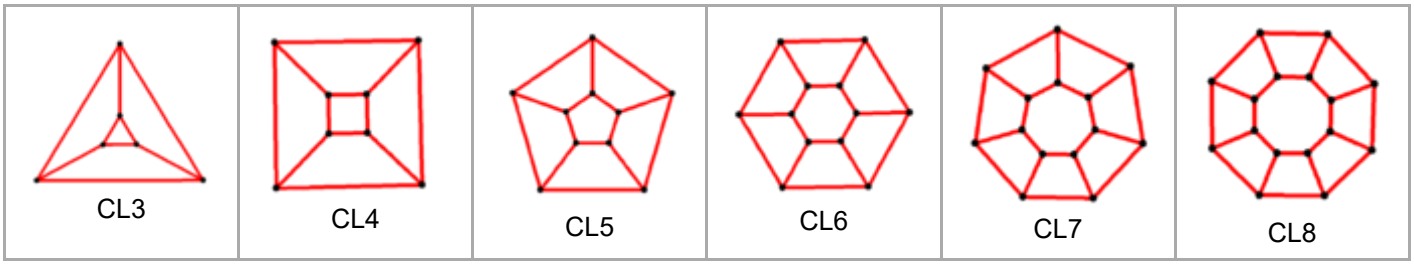
The ladder rung graphs LR_1, LR_2, LR_3, LR_4, and LR_5.

Circular ladder graph

The **circular ladder graph** CL_n is constructible by connecting the four 2-degree vertices in a *straight* way, or by the Cartesian product of a cycle of length $n \geq 3$ and an edge.[4] In symbols, $CL_n = C_n \times P_2$. It has $2n$ nodes and $3n$ edges. Like the ladder graph, it is connected, planar and Hamiltonian, but it is bipartite if and only if n is even.

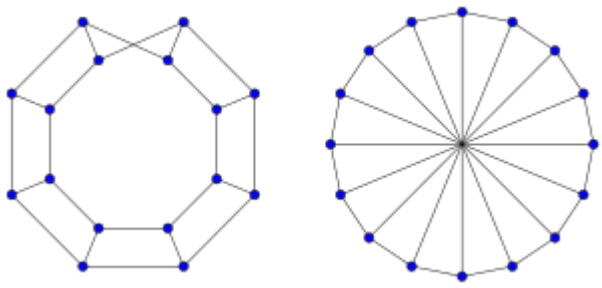
Circular ladder graph are the polyhedral graphs of prisms, so they are more commonly called **prism graphs**.

Circular ladder graphs:



Möbius ladder

Connecting the four 2-degree vertices *crosswise* creates a cubic graph called a Möbius ladder.



Two views of the Möbius ladder M_{16} .

References

1. Weisstein, Eric W. "Ladder Graph" (https://mathworld.wolfram.com/LadderGraph.html). *MathWorld*.
2. Hosoya, H. and Harary, F. "On the Matching Properties of Three Fence Graphs." J. Math. Chem. 12, 211-218, 1993.

3. Noy, M. and Ribó, A. "Recursively Constructible Families of Graphs." Adv. Appl. Math. 32, 350-363, 2004.
4. Chen, Yichao; Gross, Jonathan L.; Mansour, Toufik (September 2013). "Total Embedding Distributions of Circular Ladders". *Journal of Graph Theory*. **74** (1): 32–57. CiteSeerX 10.1.1.297.2183 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.297.2183). doi:10.1002/jgt.21690 (https://doi.org/10.1002%2Fjgt.21690).

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